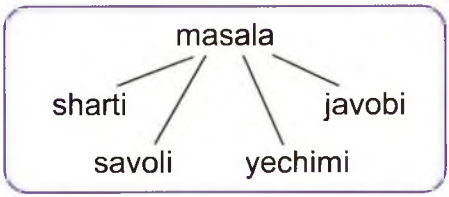
Matematika – bu nima?

Salom! Darslikning uchinchi qismida biz matematikaning asosiy tushunchasi: masalalar bilan tanishamiz. Masalalar turlarini farqlashni va ularga doir ifodalar tuzishni o‘rganamiz. Biz mantiqiy masalalar bilan tanishamiz va bunday masalalar kundalik hayotimizda qarorlar qabul qilishimizda qanday yordam berishini tushunib yetamiz.



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75. Masalaning tarkibi





2	7	6
		1
4		



Masala bu — shartga va savolga ega bo‘lgan matn.
Masala sharti bu — masalani yechish uchun berilgan ma’lumotlar.
Masala savoli bu — topish kerak bo‘lgan ko‘rsatma.
Masalaning yechimi bu — masalaning sharti bo‘yicha savolga javob topish uchun tuzilgan matematik ifoda.

1. Masala tuzing. Savolga javob bering.

Bor edi	Keltirildi	Bo‘ldi
24	12	?



2. Masalaning sharti va unga mos kelgan savolni birlashtiring.

Maktabda 33 ta katta va 22 ta kichik xona bor.	O‘g‘il bolalar qiz bolalardan nechta ko‘p?
Sinfda 12 nafar qiz bola va 23 nafar o‘g‘il bola bor.	Likopchada nechta nok bor?
Likopchada 12 ta olma va olmalardan 7 ta kam nok bor.	Maktabda hammasi bo‘lib nechta xona bor?

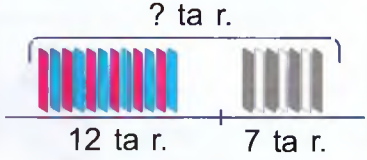
3. Sinfdoshingiz bilan musobaqalashing. Sonli zanjirnikim tezroq to‘ldiradi?



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4. Masalaning yechimini bilib oling.

O‘qituvchi darsga 12 ta rangli va 7 ta oq-qora rasm tayyorladi. O‘qituvchi darsga nechta rasm tayyorlagan?

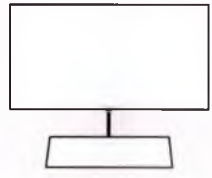


12 + 7 = 19 (rasm)
Javob: 19 ta rasm

5. 4- masala modeliga muvofiq masalani yeching.

O‘quvchilar 4 ta masala va 12 ta misol yechdilar. Ular hammasi bo‘lib nechta topshiriq bajardilar?

6. Qalamni qog‘ozdan uzmagan va chiziqlar ustidan ikki marta yurmagan holda shaklning ustidan chizib chiqing.



7. So‘zni taxmin qiling. Bu nima?

U	26 + 31	T	68 – 31
B	45 + 54	K	96 – 54
N	73 + 14	O	86 – 32

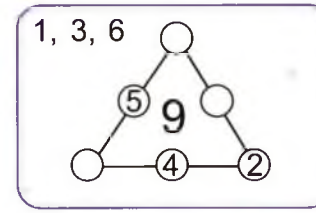
87	54	57	37	99	57	42

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76. Masalani yechish algoritmi

Masalani yechish rejasi:
1. Ma'lumotlarni tanlang
2. Savolni aniqlang
3. Ifoda tuzing







Isfalgan masalani yechish uchun reja tuzib olinadi.

1. Masalani yeching.

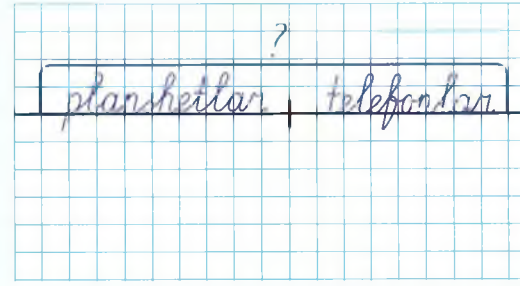


— 37 ta



— 42 ta

Hammasi ?



2. So'zni toping. Bu so'z masalani yechishdagi amallar ketma-ketligini anglatadi.

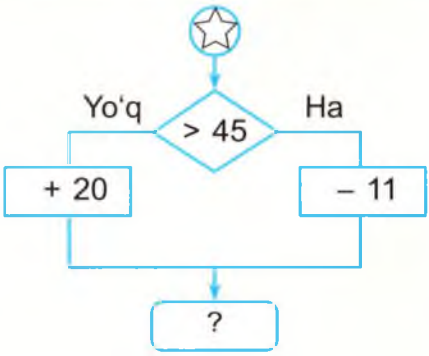
A	49+(11+9)	M	60+(56−16)
G	37−(45−22)	O	96−(64+22)
I	56+(32−11)	R	32+(33+31)
L	85+(86−74)	T	21+(78−34)

69	97	14	10	96	77	65	100

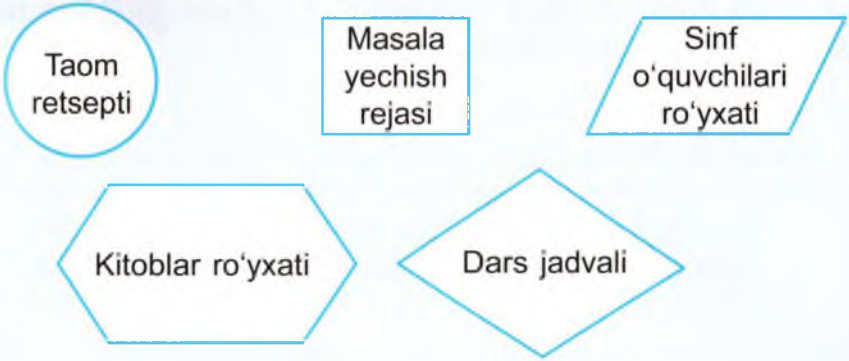
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3. Algoritm bajarilgandan so‘ng yulduzchalardagi sonlardan nima hosil bo‘ladi.





4. Algoritmga ega bo‘lgan shakllarni bo‘yang.

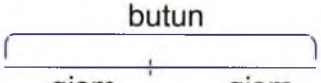


5. To‘la va bo‘sh stakanlar navbat bilan joylashsin. Faqat bitta stakanni qo‘lga olishga ruxsat beriladi.

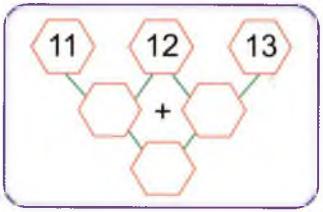


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Masala uchun sxema





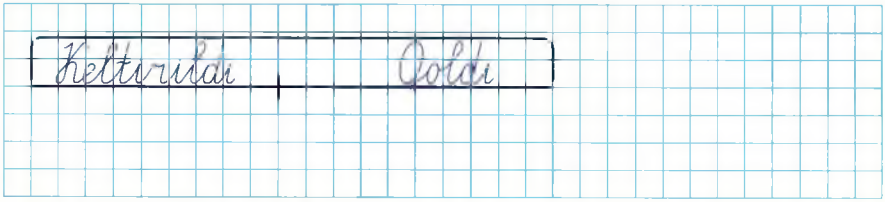




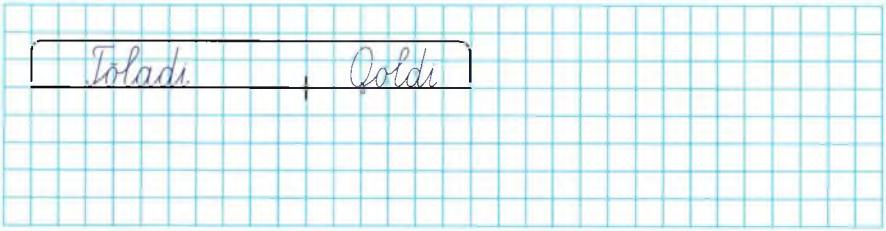
Masalani yechish rejasi algoritm deyiladi.

1) berilganlarni aniqlash;
2) masalaga savol tanlash;
3) ifoda tuzish;
4) uning qiymatini hisoblash;
5) javobini yozish.

1. 88 ta yangi o‘qish darsligidan 30 tasi keltirildi. Yana nechta kitob keltirilishi kerak?



2. Sayohatga 36 nafar o‘quvchi borishi kerak edi. 24 nafar o‘quvchining ota-onasi chiptaga pul to‘ladi. Yana nechta ota-ona pul to‘lashi kerak?



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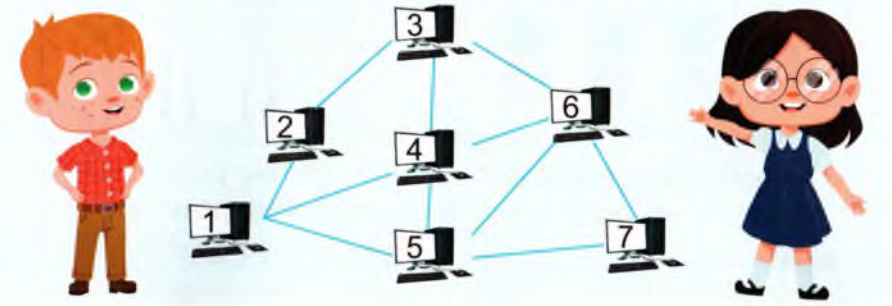
3. Taqqoslang.

20 + 34 ☐ 44	25 + 70 ☐ 70 + 23
17 + 22 ☐ 40	33 + 66 ☐ 22 + 77

4. Xatjild ustidagi ifodaning javobi yozilgan pochta qutisiga tushishi kerak. Barcha qutilarga o‘z konvertlarini soling.



5. Qog‘ozdan qalamni uzmagan hamda bitta chiziq ustidan ikkinchi marta o‘tmagan holda havorang chiziqlar ustidan chizib boring. Yo‘lingizda qaysi raqamli kompyuterdan o‘tgan bo‘lsangiz, shu raqamlarni tartib bilan yozing.



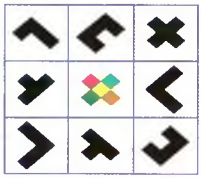
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78. Masala shartining berilish usullari

matn
rasm
qisqacha sharti
sxema



Shakl qaysi soya beradi?



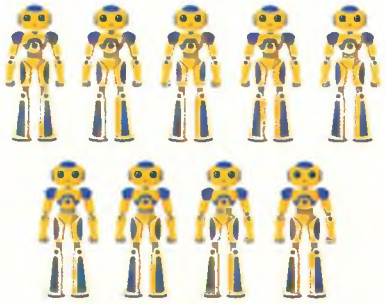
Masala sharti matn, rasm yoki sxema ko‘rinishida berilishi mumkin.



1. Matn.
Abror 12 ta o‘quv filmini tomosha qildi.
Uning tomosha qilish uchun 24 ta filmi bor.
Abrorda jami nechta ta’limiy film bor?

2. Rasm.





Robotlar jami nechta?

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3. Qisqacha sharti.

Avval — 8 bet
Qoldi — 3 bet
? bet

4. Sxema.

?
Radio 5 , Televizor 10

5. Har bir juftlikda ifodalar qanday qoida bo'yicha tuzilgan? Ularning qiymatini hisoblang.

a) 23 + 6
26 + 3

b) 45 + 52
42 + 55

d) 17 + 21
11 + 27

e) 34 + 43
33 + 44

6. Sinfdoshingiz bilan musobaqalashing. Kim tezroq sehrli kvadratni to'ldiradi? Ustun va qatorlardagi sonlar yig'indisi 15 ga teng bo'lishi kerak.

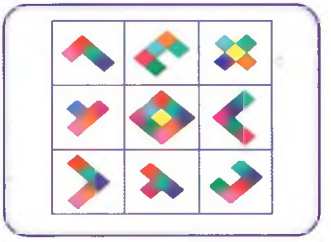
	9	
3		7
8		

8		
	5	
6		2

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butun, qismlar
masalalar yechish
ifodalar





1. Agar bitta robot mashinaga 46 qop, ikkinchisi 42 qop chiqindi yuklasa, mashina necha qop chiqindini shahardan olib chiqadi?



2. Misollarni yeching. Javoblarni kamayish tartibida joylashtiring. Harflarni yonma-yon yozing, shunda siz odamga o'xshash robotning nomini bilib olasiz.

A	53 + 41	D	40 − 30	N	97 − 16
O	28 + 21	D	20 + 50	R	93 − 32
I	14 + 32	Javob:			

3. Rasmga ko'ra masala o'ylab toping, ifoda tuzing va uning qiymatini toping.



31



23



25

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4. Ko'rgazmada robotlar to'plagan ballar jadvalga yozilgan. Eng katta umumiy ball to'plagan robot g'olib sanaladi.

		 Bibi	 Pinki	 Aybo
1- topshiriq	5	4	5	3
2- topshiriq	3	4	2	5
3- topshiriq	5	2	4	4
Umumiy ball				

Musobaqada qaysi robot g'olib bo'ldi?

5. Sinfdoshingiz bilan musobaqalashing. Kim birinchi bo'lib o'zining robotini yonayotgan uyga olib o'tadi?
Kataklardan shunday yurish kerakki, misollarning birinchi soni oldin kelgan misolning natijasiga teng bo'lsin.

88 – 14	74 + 11		73 + 12	30 + 43
30 + 58	22 + 52	32 + 15	40 + 33	57 – 27
49 – 19	24 + 25	45 – 13	31 + 11	42 + 15
40 + 9	36 – 12	48 + 11	56 – 25	30 + 12
20 + 10	30 + 6	24 – 14	40 + 16	30 + 10





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butun va qismlar

qism + qism = butun





1. Masalaga ifoda tuzing va uni yeching.

Harbiy dronlar



42



Fuqorolik dronlari

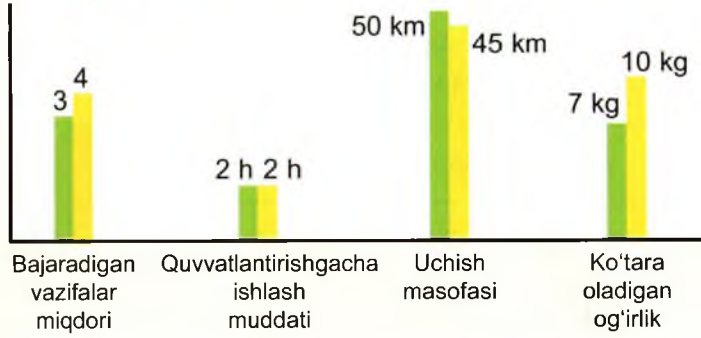
27

? ta dron

2. Kichik loyiha. Fermer o‘z dalalarini kuzatib turish uchun dron sotib olishni rejalashtirdi. Fermerga eng yaxshi dronni tanlashiga yordam bering.

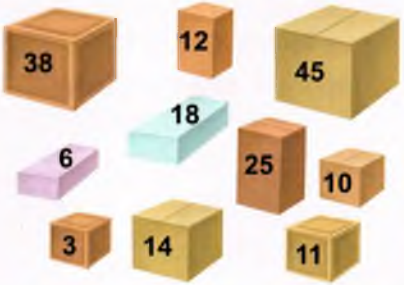






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3. Katta dronlar 30 kg gacha bo‘lgan yuklarni yetkazib bera oladi. Bunday dron ko‘tara oladigan og‘irlikdagi yuklarni umumiy chiziq bilan aylanaga oling.



4. Hisoblang.

34 + 11	47 + 21	23 + 35
53 − 11	18 − 16	98 − 56

5. Masalani yeching.

Birinchi dron – 35 ta foto
Ikkinchi dron – 49 ta foto
} ?

6. O‘rdakchaning og‘irligini toping.

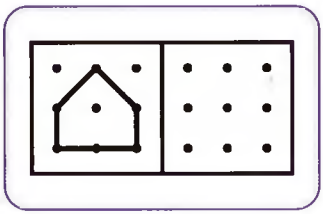


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81. Butunni topishga doir masalalar

butun va qismlar
qism + qism = butun





1. Hisoblang.

7+4	6+6	3+9
8+5	9+7	8+8
6+9	7+7	7+8

2. Masala shartini qisqacha ifodalang va yeching.
Uy ishlariga yordam beruvchi robotni yaratish ustida 12 nafar muhandis va 8 nafar dasturchi ishladi. Hammasi bo'lib robotni necha kishi yaratdi?

Muhandis –
Dasturchi –
] ?

3. Sxema bo'yicha to'rtta masala tuzing. Masalalarga ifoda tuzing va ularning qiymatlarini toping. Masalalarning umumiyligi nimada? Nima bilan farq qiladi?

Hammasi bo'lib?
Yig'ib olindi ; Qoldi

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4. Eng yaxshi robot-changyutgichni tanlashga yordam bering. Har bir xossasi talab darajasida ekanligiga ahamiyat bering va shu katakni bo‘yang. Qaysi birida bo‘yalgan kataklar ko‘p? Eng yaxshisini qalam bilan aylanaga oling.

	 „Aqlli robot“	 „Uy bekasi roboti“
Chang tortish quvvati	100	85
Ishlash vaqti	2 soat	2 soat
Ovoz chiqarib ishlash darajasi	45	50
Robotning balandligi	6 santimetr	7 santimetr
To‘siqlarni ko‘rish xossasi	+	+
Tozalash (suv bilan)	+	–
Kafolat	2 yil	1 yil

5. Qanday qilib yig‘indining qiymatini tezroq hisoblash mumkin ekanini o‘ylab toping.

1 + 3 + 5 + 7 + 9

1 + 2 + 3 + 7 + 8 + 9

1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9

6. 2, 3, 6, 8, 9 raqamlarining hammasidan foydalanib, tenglik tuzing.

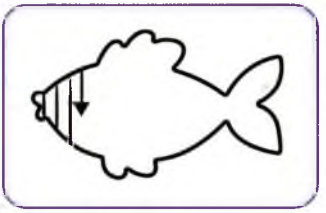
FARG‘ONA VILOYATI
RISHTON TUMANI
MAKTABGACHA VA MAKTAB
TA’LIMI BO‘LIMI
33-UMUMIY
O‘RTA TA’LIM
MAKTABI
№ 40
2023

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82. Butunni topishga doir masalalar

Butun va qismlar
qism + qism = butun





=

1. Sonlarni 10 gacha to'ldiring.

9 + □ = 10	5 + □ + 4 = 10	□ + 8 = 10
□ + 5 = 10	2 + 3 + □ = 10	□ + 1 = 10

2. Hisoblang.

7 + 3 = □□	5 + 6 = □□	9 + 4 = □□	7 + 9 = □□
6 + 4 = □□	8 + 7 = □□	5 + 8 = □□	9 + 8 = □□

3. Kasalxonada nechta robot ishlayapti?



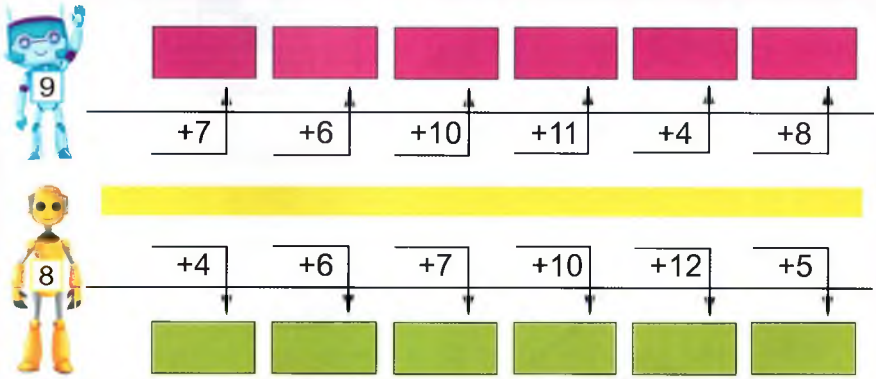


4. Kasalxonaning ikkinchi qavatida bir necha bemor davolanyapti. Ulardan yetti nafari dorilarini olishdi. Qolgan to'rt nafar bemorga dorilarni robot olib boradi. Kasalxonaning ikkinchi qavatida necha nafar bemor davolanyapti?



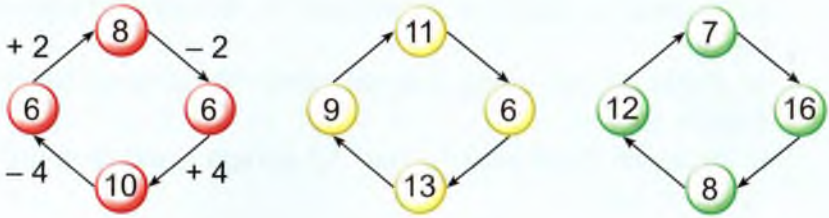
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5. Sinfdoshingiz bilan musobaqalashing. Yo'lakdan robotni kim tezroq olib o'tadi? Kim to'g'ri bajardi?



6. Har bir strelkaning ustida qanday son va qanday amal belgisi turishi kerak? Namunada topshiriq qanday bajarilganini ko'ring.

Namuna:



7. a) 1 dan 20 gacha bo'lgan sonlarni ortib borish tartibida ayting.

b) 2 dan 20 gacha bo'lgan juft sonlarni ayting.

d) 5 dan 30 gacha bo'lgan sonlarni beshtalab sanang.

e) 10 dan 100 gacha bo'lgan sonlarni o'ntalab sanang.

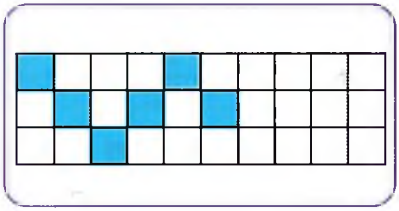
f) 5 dan 15 gacha bo'lgan sonlarni teskari tartibda sanang.

g) 99 sonining „qo'shnilarini" ayting.

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Butun va qismlar
qism + qism = butun
butun – qism = qism





1. Ushbu qismlarning bir butunligi nimadan iboratligini toping.





2. a) Birinchi qo‘shiluvchi 7, ikkinchisi 13. Yig‘indini toping.
b) Qismlari 6 va 12 sonlaridan iborat bo‘lgan butunni toping.
d) Yig‘indi 13, ikkinchi qo‘shiluvchi 4. Birinchi qo‘shiluvchini toping.
e) Butun 17 ga, uning ikkinchi qismi 10 ga teng. Birinchi qismini toping.
f) 20 sonini hosil qilish uchun 17 soniga qanday sonni qo‘shish kerak?
g) 19 sonini hosil qilish uchun qanday sonni 13 ga qo‘shish kerak?

3. Masalani yeching.
Robototexnika mashg‘uloti uchun 46 ta qism tayyorlandi. Ushbu qismlarning 36 tasidan robot yasaldi. Nechta qism qoldi?

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4. Bo‘sh katakchalarni to‘ldiring.

40 + □□ = 68	12 + □□ = 46	53 + □□ = 60
□□ + 34 = 98	□□ + 36 = 76	□□ + 73 = 80

5. So‘roq belgilari ortida nima yashiringan?

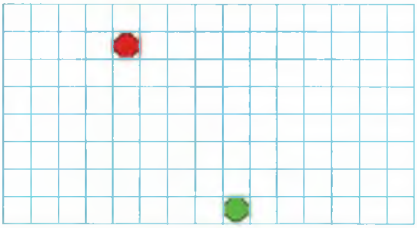
Bunda 24 ta egri chiziq + ? ta siniq chiziq = 69 ta chiziq

17 ta uchburchak + ? kvadrat = 79 ta shakl bo‘lsa-chi?

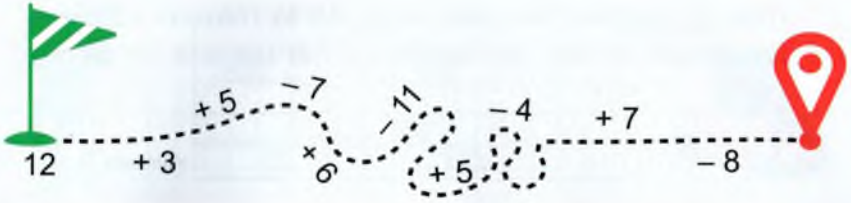
62 ta konus + ? silindr = 89 ta hajmli jism bo‘lsa-chi?

6. Robot qizil doirachadan yashil doirachagacha bosib o‘tgan yo‘lining algoritmini belgilang.

a) 1→, 3↓, 3←, 3↓, 2→
b) 1→, 3↓, 4←, 2↓, 1→
d) 1→, 2↓, 2→, 4↓, 1→
e) 1→, 2↓, 2→, 1↓, 1→



7. Topshiriqni bajarib bo‘lgandan so‘ng, robot 9 sonini hosil qildi. U topshiriqni to‘g‘ri bajarganmi? Tekshiring.

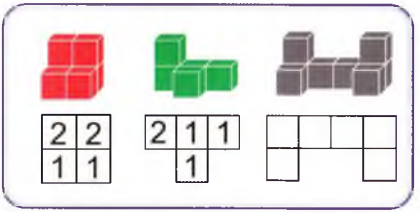


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84. Qismlarni topishga doir masalalar

Butun va qismlar
qism + qism = butun
butun – qism = qism





1. Jadvalni to‘ldiring.

Qo‘shiluvchi	13	42		11	98		
Qo‘shiluvchi	54		13		1	60	40
Yig‘indi		68	57	21		100	40

2. Toping.

17 soni hosil bo‘lishi uchun 13 soniga nechta birlik yetmaydi?

90 soni hosil bo‘lishi uchun 50 soniga qaysi sonni qo‘shish kerak?

56 soni hosil bo‘lishi uchun 53 sonini nechtaga orttirish kerak?

3. Butun Jahon robototexnika kuniga bag‘ishlangan bayramda 18 ta to‘garak qatnashchisi va ularning ota-onalariga maxsus belgilar tarqatildi. Agar 48 ta maxsus belgi tarqatilgan bo‘lsa, bayramga nechta ota-ona kelgan?

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4. Taqvimdan foydalanib, masalani yeching.
Aziz butun Jahon robototexnika kuniga bag'ishlangan bayramni nishonlash uchun 17 kun qolganini taqvimga qarab hisobladi. U hisobni qaysi sanadan boshlagan?

Yanvar / Fevral

Dushanba	Seshanba	Chorshanba	Payshanba	Juma	Shanba	Yakshanba
30	31	1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31	1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	1



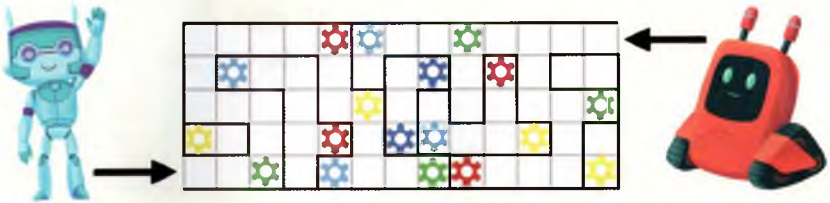
5. Ortiqcha ifodani ko'rsating.

a) $8-3$ $5+3$ $8-5$ $5-3$

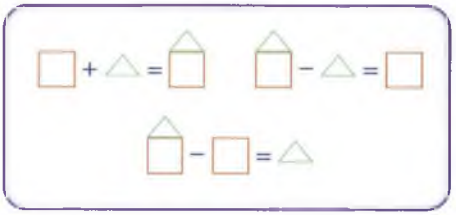
b) $9+4$ $13-9$ $4+1$ $13-4$

d) $6+14$ $14-6$ $20-6$ $20-14$

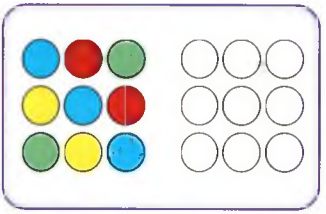
6. Sinfdoshingiz bilan musobaqalashing. Kimning roboti ko'proq detal yig'adi? Labirintda nechta detal qoladi?



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1. Rasmda amallarni ifodalang.

6 + 7



17 − 8



2. Masalalarni yeching.

Jo'natma Yevropadan O'zbekistonga 5 kun davomida samolyotda va yana 5 kun poyezdda keltirildi. Jo'natma necha kunda kelgan?

Pochtachi 78 ta xonadonga xatlarni yetkazishi kerak. Agar u 32 ta xonadonga xatlarni yetkazgan bo'lsa, yana nechta xonadon qoldi?

3. Noto'g'ri turgan qutilar ustiga chiziq torting.

30 dan kichik bo'lgan



45 dan katta bo'lgan



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4. Rasmda pochtadan uygacha bo'lgan yo'l tasvirlangan. Chiziqlar yonida yozilgan raqamlar shu masofani bosib o'tishga sarflangan vaqtni anglatadi. Pochtachining uyiga tezroq yetib borishiga yordam bering. Eng qisqa yo'lni bo'yang.



5. Pochtachi havorang, sariq va yashil ranglardagi 3 ta xat olib keldi. Xatning rangi eshik rangiga mos kelmaydi. Sariq xatni o'rtadagi eshikka olib kelmadi. Darchalarda qoldirilgan xatlarning rangini tasvirlang.



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